PROJECT BRIEF / FULL-STACK ENGINEERING

CentraLog

Connecting asset custody with maintenance and valuation.

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PROBLEM & USERS

Institutions need to know where equipment is, who holds it, and whether it is in use or under repair. CentraLog connects physical tracking with a valuation calculation that uses maintenance history.

Built for: Inventory staff · managers · accountants · administrators

MY CONTRIBUTION

I built CentraLog independently as a school project, implementing the asset lifecycle, backend services, frontend workflows, and valuation strategies.

MAIN TECHNOLOGIES

ASP.NET Core 8 / C# / React 19 / TypeScript / EF Core 8 / MySQL / MariaDB / JWT / Vitest / xUnit

WORKFLOW

01. Register and activate

02. Move with a record

03. Track maintenance

04. Calculate and report

SYSTEM STRUCTURE

Interface: React / TypeScript

API: ASP.NET Core / JWT

Domain: Depreciation strategies / worker

Persistence: EF Core / MySQL or MariaDB

ENGINEERING DECISION

Use interchangeable depreciation strategies to calculate values from asset and maintenance records on read. Group multi-asset mutations and audit entries inside EF Core transactions.

TRADE-OFF

Calculating on read adds computation but avoids maintaining a separate cached valuation. Repair windows are handled through the application's explicitly defined downtime-adjustment policy.

READ CASE STUDY | TRY WALKTHROUGH | VIEW SOURCE

Portfolio case studies and walkthroughs provide further detail and sample interactions.